SPRING **MID SEMESTER EXAMINATION-2025**

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Cloud Computing
[CS 30010]

Time: 1 1/2 Hours **Full Mark: 20**

Answer Any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions. [1 Mark X 5]

   a) Differentiate between single-tenant and multi-tenant architecture in cloud environment.
   b) What is hardware virtualization and how many types of hardware virtualization are there?
   c) What are the implications of cloud computing on the traditional IT infrastructure?
   d) What are the NIST characteristics of cloud computing?
   e) What are the characters existing in the cloud computing environment?

2. (A) What is Server Consolidation? Describe the architecture of server consolidation with suitable diagram. [2.5 Marks]

   (B) Explain the early technologies that played a significant role in the evolution of cloud computing and how they influenced the development of businesses over time. Give the advantages and disadvantages of each technology. [2.5 Marks]

3. KIIT university is planning to develop an online platform for delivering virtual classes, managing student records, and providing additional resources like lecture materials and assignments. The platform needs to be scalable to handle a large number of students, secure to protect sensitive data like student records, and reliable with minimal downtime. Additionally, the university has limited IT staff and infrastructure resources. As a cloud computing expert, recommend the appropriate cloud service model (IaaS, PaaS, or SaaS) and the appropriate cloud deployment model (Public Cloud, Private Cloud, Hybrid Cloud, or Community Cloud) for this scenario. Justify your choices by explaining how your recommendations meet the university's needs for scalability, security, reliability, and ease of management. [5 Marks]

4. (A) What are the key components of a cloud-based data centre? What are the different types of modular portable data centre? [2.5 Marks]

   (B) What is resource replication and how it is important in the cloud environment? [2.5 Marks]

5. (A) Write a short note on logical network perimeter and what are the components of the same? [2.5 Marks]

   (B) What are the steps to host a web service in cloud environment? [2.5 Marks]

***** Best of Luck *****